LABORATORY PROGRAM – 3

Perform the following DB operations using

Cassandra Questions:

a) Create a keyspace by name Library
b) Create a column family by name Library-Info with attributes
   - Stud_Id Primary Key,
   - Counter_value of type Counter,
   - Stud_Name, Book-Name, Book-Id,
   - Date_of_issue
c) Insert the values into the table in batch
d) Display the details of the table created and increase the value of the counter
e) Write a query to show that a student with id 112 has taken a book “BDA” 2 times.
f) Export the created column to a csv file
g) Import a given csv dataset from local file system into Cassandra column family

```
cqlsh> CREATE KEYSPACE Library
   ... WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> USE Library;
cqlsh:library> CREATE TABLE Library_Info (
           ...     Stud_Id INT PRIMARY KEY,
           ...     Stud_Name TEXT,
           ...     Book_Name TEXT,
           ...     Book_Id INT,
           ...     Date_of_issue DATE
           ... );
cqlsh:library> CREATE TABLE Book_Counter (
           ...     Stud_Id INT,
           ...     Book_Name TEXT,
           ...     Counter_Value COUNTER,
           ...     PRIMARY KEY (Stud_Id, Book_Name)
           ... );
cqlsh:library> BEGIN BATCH
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (111, 'Manoj', 'BDA', 201, '2024-04-01');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (112, 'Rahul', 'BDA', 202, '2024-04-02');
           ...
           ... INSERT INTO Library_Info (Stud_Id, Stud_Name, Book_Name, Book_Id, Date_of_issue)
           ... VALUES (113, 'Anjali', 'DBMS', 203, '2024-04-03');
           ... APPLY BATCH;
cqlsh:library> UPDATE Book_Counter
           ... SET Counter_Value = Counter_Value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
cqlsh:library> UPDATE Book_Counter
           ... SET Counter_Value = Counter_Value + 1
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';
cqlsh:library>
```

```
cqlsh:library> SELECT * FROM Library_Info;

 stud_id | book_id | book_name | date_of_issue | stud_name
---------+---------+-----------+---------------+----------
     111 |     201 |       BDA |    2024-04-01 |     Manoj
     113 |     203 |      DBMS |    2024-04-03 |    Anjali
     112 |     202 |       BDA |    2024-04-02 |     Rahul

(3 rows)
cqlsh:library> SELECT * FROM Book_Counter;

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2

(1 rows)
cqlsh:library> SELECT * FROM Book_Counter
           ... WHERE Stud_Id = 112 AND Book_Name = 'BDA';

 stud_id | book_name | counter_value
---------+-----------+---------------
     112 |       BDA |             2

(1 rows)
cqlsh:library> cqlsh -e "COPY Library.Library_Info TO 'library.csv' WITH HEADER = TRUE;"
           ...
cqlsh:library>
```

```
┌──(thilak㉿kali)-[~]
└─$ cqlsh -e "COPY Library.Library_Info TO 'library.csv' WITH HEADER = TRUE;"
/usr/bin/cqlsh.py:473: DeprecationWarning: Legacy execution parameters will be removed in 4.0. Consider using execution profil
es.
/usr/bin/cqlsh.py:503: DeprecationWarning: Setting the consistency level at the session level will be removed in 4.0. Consider
 using execution profiles and setting the desired consistency level to the EXEC_PROFILE_DEFAULT profile.
Using 5 child processes
/usr/bin/cqlsh.py:1555: DeprecationWarning: setDaemon() is deprecated, set the daemon attribute instead

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 3 rows; Rate:      12 rows/s; Avg. rate:      12 rows/s
3 rows exported to 1 files in 0.279 seconds.

┌──(thilak㉿kali)-[~]
└─$ cqlsh -e "COPY Library.Library_Info FROM 'library.csv' WITH HEADER = TRUE;"
/usr/bin/cqlsh.py:473: DeprecationWarning: Legacy execution parameters will be removed in 4.0. Consider using execution profil
es.
/usr/bin/cqlsh.py:503: DeprecationWarning: Setting the consistency level at the session level will be removed in 4.0. Consider
 using execution profiles and setting the desired consistency level to the EXEC_PROFILE_DEFAULT profile.
Using 5 child processes
/usr/bin/cqlsh.py:1553: DeprecationWarning: setDaemon() is deprecated, set the daemon attribute instead

Starting copy of library.library_info with columns [stud_id, book_id, book_name, date_of_issue, stud_name].
Processed: 3 rows; Rate:       4 rows/s; Avg. rate:       6 rows/s
3 rows imported from 1 files in 0.482 seconds (0 skipped).

┌──(thilak㉿kali)-[~]
└─$
```